Hidden Currents: A Peak-Stress Water Index for Auditing AI Data Center Sustainability and Informing Disclosure Policy

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Data and code: https://github.com/hrihaan19/pswi-data-center-audit
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***Abstract*—Data center water use is usually reported as Water Usage Effectiveness (WUE), an annual, fleet-averaged ratio of water to IT energy. WUE hides two dimensions of impact on host communities: the concentration of demand on peak summer days and the scarcity of water where a facility is located. Prior work has examined these dimensions separately. We propose the *Peak-Stress Water Index* (PSWI), the product of annual WUE, a peaking factor β (peak-day to average-day water use), and the facility's Baseline Water Stress (BWS) from WRI Aqueduct 4.0. We apply PSWI to an open dataset of 48 hyperscale and colocation facilities run by seven operators in North America, Europe, and Asia, using only WUE values that operators publish or that can be derived from their published totals. Because none of these operators publishes facility-level WUE, facilities inherit fleet or regional values, so PSWI isolates how location changes the burden behind the same headline number. PSWI spans a factor of about 1,500 across facilities; within one operator that applies a single fleet WUE, it differs 267-fold between Henderson, Nevada and Hamina, Finland. Company rankings by mean PSWI depart from rankings by operator WUE (Spearman $\rho = 0.71$), although bootstrap intervals for individual company shifts include zero. Our climate-based β is strongly correlated with BWS ($r = 0.84$); removing it leaves facility and company rankings nearly unchanged ($\rho = 0.96$ for both), so the results are driven mainly by location stress. In an idealized reallocation exercise, moving 10% of workload from the highest- to the lowest-PSWI facilities lowers aggregate PSWI by 35% (95% interval 21–46% under assumed input uncertainty), showing that impact is concentrated in a small tail. None of the audited operators publishes peak or monthly water data. We outline a facility-level disclosure framework that borrows the phase-in architecture of U.S. securities disclosure rules but could be implemented through state and local water permitting. Data and code are openly released.**

***Index Terms*—data centers, sustainable computing, water stress, water usage effectiveness, AI infrastructure, environmental disclosure, sensitivity analysis**

I. Introduction

The rapid expansion of artificial intelligence has been accompanied by a rapid expansion of the data center infrastructure that powers it. Recent projections place U.S. data center electricity consumption at 6.7% to 12.0% of national totals by 2028 [1], and the International Energy Agency projects global data center electricity demand to roughly double by 2030 [2]. Water demand is growing alongside. Han et al. estimate that, if 2024 water-use intensity persists, U.S. data centers could require 697 to 1,451 million gallons per day of new public water capacity through 2030—comparable to New York City's average daily supply of roughly 1,000 million gallons—with a total valuation on the order of $10 billion and up to $58 billion in a high-growth case [3].

The industry's primary water metric, Water Usage Effectiveness (WUE) [4], divides annual site water use by annual IT energy. It is usually reported as a single fleet-wide average [5]–[7], and it treats every liter as equivalent. This obscures two dimensions of impact:

1) **Temporal concentration of demand.** Han et al. [3] show that data center water demand is concentrated on the hottest days of the year, with daily peaking factors estimated at roughly 3 to 10 depending on climate and cooling configuration. Public water systems are sized for peak demand, so a low annual WUE can coexist with heavy peak-day draws.
2) **Spatial heterogeneity of stress.** A liter consumed in the Phoenix area, in a Colorado River basin that has experienced persistent drought since about 2000, does not carry the same consequences as a liter at Hamina, Finland, where Google cools with seawater. Wu et al.'s SCARF framework [8] weights water consumption by a location-specific stress factor derived from the World Resources Institute's Aqueduct dataset [9].

These dimensions have been studied largely separately: Han et al. analyze peak demand and system capacity without weighting by basin-level water stress, while Wu et al. weight by stress without modeling peak-day concentration. To our knowledge, no published metric combines them.

A. Contributions

1) **A composite metric.** We define the *Peak-Stress Water Index* (PSWI) as the product of annual WUE, a peaking factor, and Baseline Water Stress, combining the peaking framing of Han et al. with the stress weighting of Wu et al. (Section IV).
2) **An auditable dataset with documented provenance.** We assemble 48 facilities operated by seven providers.

Every WUE value is either published by the operator or derived arithmetically from published totals, and is labeled accordingly (Section V).

3) **Findings with explicit robustness tests.** We document a roughly 1,500-fold range in PSWI, a 267-fold difference between two facilities of one operator sharing a single fleet WUE, and departures between WUE-based and PSWI-based company rankings. We test these against perturbation and removal of the peaking factor, bootstrap resampling, and inclusion of indirect (Scope 2) water (Section VI).
4) **A disclosure audit and policy framework.** None of the seven operators publishes peak or monthly water data. We propose facility-level disclosure requirements and discuss implementation in light of the history of the U.S. Securities and Exchange Commission's climate disclosure rules (Section VII).

II. Background

A. Data Center Water Use

Data centers use water directly, mostly for on-site cooling, and indirectly through electricity generation [3], [10]. Evaporative cooling removes heat efficiently by evaporating water; dry cooling avoids on-site water use but generally requires more electricity, especially on hot days, so the two resources trade off [3], [11]. Indirect (Scope 2) water use often exceeds direct use in volume [10], but it is drawn at power plants, frequently in other basins and largely outside public drinking-water systems. We focus on direct use and test sensitivity to Scope 2 in Section VI-F.

B. The WUE Metric and Its Limitations

The Green Grid defines WUE as [4]

$$\text{WUE} = \frac{\text{Annual site water usage (L)}}{\text{Annual IT equipment energy (kWh)}}. \tag{1}$$

As used in corporate reporting, WUE has three limitations for assessing local impact. **Annualization** erases the peak signal that matters for systems sized to peak demand. **Location-blindness** treats a liter in a drought-stressed basin like a liter drawn from the Baltic. **Aggregation** into fleet or regional averages dilutes high-impact sites with low-impact ones. Operators also define the numerator differently—some use water withdrawal, others consumption—which limits comparability across companies [6].

III. Related Work

a) Sustainable AI and computing: Strubell et al. [12] documented the energy footprint of training large language models; Patterson et al. [13] extended this to carbon. Wu et al. [14] provided an industry perspective from Meta, Schwartz et al. [15] introduced the Green AI framing, and Luccioni et al. [16] performed a life-cycle carbon analysis of BLOOM. Most of this literature treats water as secondary to carbon or omits it.

b) Geographical load balancing: Islam et al. [17] exploited spatio-temporal diversity to reduce water use in geo-distributed data centers. Acun et al. [18] introduced Carbon Explorer, Radovanović et al. [19] described Google's carbon-aware computing system, and Hanafy et al. [20] quantified the cost of reducing cloud carbon emissions. Li et al. [21] proposed equity-aware geographical load balancing, Liu et al. [22] studied geographical server relocation, and Bian et al. [23] applied carbon-aware federated learning. In this line of work, water is usually treated through unweighted volumes.

c) Water-aware computing: Wu et al.'s SCARF framework [8] defines the Adjusted Water Impact (AWI), which multiplies on-site and off-site water consumption by a water stress factor derived from Aqueduct [9]. Jiang et al.'s WaterWise [24] co-optimizes carbon and water footprints in scheduling, and ThirstyFLOPS [25] models the water footprint of HPC systems. Han et al. [3] quantify data centers' peak demands on U.S. public water systems, estimate daily peaking factors, and argue that available peak capacity rather than annual volume is the binding constraint. Li et al. [10] estimate the water footprint of AI models. De Vries-Gao [26] estimates the global water footprint of AI systems (312.5–764.6 billion liters in 2025) and reviews which environmental metrics major operators disclose; our disclosure audit (Section VI-H) is narrower, asking whether the specific inputs needed for a facility-level PSWI are published. Siddik et al. [27] estimate the environmental footprint of U.S. data centers, and Karimi et al. [11] examine water–energy trade-offs in hot-arid climates.

d) Our contribution: PSWI combines the peak and stress dimensions in a single facility-level index and applies it to a cross-operator dataset whose inputs are fully traceable to public documents.

IV. The PSWI Metric

A. Definition

For facility i,

$$\text{PSWI}_i \;=\; \text{pWUE}_i \cdot \text{BWS}_i \;=\; \text{WUE}_i \cdot \beta_i \cdot \text{BWS}_i, \tag{2}$$

where WUE_i is annual WUE in L/kWh; $\beta_i \geq 1$ is the daily peaking factor, the ratio of peak-day to average-day water use; and BWS_i is Aqueduct 4.0's baseline water stress, the dimensionless ratio of total water demand to available renewable supply [9]. We write $\text{pWUE}_i = \text{WUE}_i \cdot \beta_i$ for the peak-day water intensity implied by the peaking factor; this notation is ours, building on the peaking-factor analysis of Han et al. [3]. PSWI has units of L/kWh and can be read as *stress-weighted peak-day water intensity*.

B. Rationale for the Multiplicative Form

Each factor scales the previous one: β converts average-day to peak-day water intensity, and BWS weights each liter by the scarcity of water where it is drawn. A product is therefore the natural composite. It reduces to stress-weighted water intensity when $\beta = 1$, the logic of Wu et al.'s AWI [8], and to peak water intensity when BWS is held fixed, the logic of Han et al. [3]. An additive index would require arbitrary

weights to combine quantities measured in different units. The multiplicative form is a modeling choice rather than an empirical finding; Section VI-E reports how much our results depend on the β component.

C. Properties

a) Monotonicity: For positive inputs, PSWI is strictly increasing in each factor with the others held fixed, since $\partial\,\mathrm{PSWI}/\partial X_k = \prod_{j\neq k} X_j > 0$ for $X_k \in \{\mathrm{WUE}, \beta, \mathrm{BWS}\}$.

b) Log-additivity: Because $\log \mathrm{PSWI} = \log \mathrm{WUE} + \log\beta + \log \mathrm{BWS}$, the cross-facility variance decomposes exactly as

$$\mathrm{Var}(\log \mathrm{PSWI}) = \sum_k \mathrm{Var}(\log X_k) + 2\sum_{j<k} \mathrm{Cov}(\log X_j, \log X_k). \tag{3}$$

c) Rank invariance under uniform rescaling: Multiplying any factor by the same constant at every facility rescales every PSWI equally and leaves rankings unchanged. Only differences *between* facilities in a factor can affect rankings.

d) Limiting behavior: As BWS $\to$ 0, PSWI $\to$ 0 regardless of water volume, reflecting negligible freshwater scarcity. When BWS > 1, demand already exceeds renewable supply and PSWI exceeds pWUE.

D. Known Limitations of the Metric

Box 1: Known limitations in this application

Assumed peaking factor. β is assigned by climate class, not measured. It is strongly correlated with BWS across our sample ($r = 0.84$ in logs), so location enters PSWI twice. Removing β entirely changes facility rankings little ($\rho = 0.96$; Section VI-E).

No facility-level WUE. No operator in our sample publishes it; each facility inherits its operator's fleet or regional value.

Water stress inputs. BWS values are annual, rounded, and assigned manually by metropolitan area.

Scope 2 excluded. Indirect water is omitted from the main specification and tested in Section VI-F.

V. Methodology

A. Sample Frame and Inclusion Criteria

We include a facility if (i) it is a hyperscale or major colocation data center; (ii) its operator publishes WUE at the fleet or regional level, or publishes fleet totals from which WUE can be derived; (iii) its location is publicly known at least to the metropolitan level; (iv) it was operating during the reporting year; and (v) it lies within Aqueduct 4.0 coverage. We excluded xAI, whose Memphis facility has drawn public attention but for which we found no published WUE, and Chinese hyperscalers, whose published water data we could not link to facilities.

The resulting purposive sample contains 48 facilities: Google (15), Microsoft (8), Meta (8), Amazon Web Services (6), Equinix (6), Digital Realty (3), and QTS (2). Thirty-two are in the United States, 11 in Europe, and 5 in Singapore and Taiwan. The sample is not statistically representative of global capacity, and we draw no inferences about operators that do not disclose.

TABLE I
WUE applied to each operator's facilities (published or derived)

Operator	WUE (L/kWh)	Year	Basis	Source
Google	0.96	2023	Derived[a]	[28]
Meta	0.19	2024	Fleet[b]	[6]
Amazon Web Services	0.18	2023	Fleet	[7]
Microsoft, Americas	0.38	FY24	Regional	[5]
Microsoft, EMEA	0.03	FY24	Regional	[5]
Microsoft, Asia-Pac.	0.03	FY24	Regional	[5]
Equinix	0.95	2024	Fleet	[29]
Digital Realty	0.59	2025	Fleet	[30]
QTS	0.82	2024	Fleet	[31]

[a] 2023 data center water consumption (6.1 billion gallons, ≈23.1 billion L) divided by data center electricity use (24 TWh). Because the denominator is total rather than IT energy, this understates Green Grid WUE by roughly the fleet's power usage effectiveness. [b] Meta computes WUE from water withdrawal.

B. WUE Assignment

Table I lists the WUE applied to each operator's facilities and its source. No operator in the sample publishes WUE for individual facilities. Google publishes facility-level water volumes but no WUE; we derive a fleet value from its published 2023 data center water consumption and electricity use. Microsoft publishes WUE by region, which we apply by facility location.

Operators define WUE inconsistently (withdrawal versus consumption; IT versus total energy) and report for different years (2023–2025). Equinix, Digital Realty, and QTS do not state units in the documents we reviewed; we assume L/kWh following The Green Grid [4]. These differences limit cross-operator comparison of WUE itself, and are one reason our central results concern variation *within* operators.

C. Water Stress Assignment

Each facility is assigned Aqueduct 4.0's baseline annual BWS [9] for its metropolitan area, recorded as the raw demand-to-supply ratio rounded to two decimals. The file `water_stress.csv` lists the basin and value for each location. Values were assigned manually rather than through a geospatial join of facility coordinates to Aqueduct catchments, and Singapore uses a single national value.

D. Peaking Factor Assignment

Operators do not disclose peaking factors. We assign β by climate class: cool 1.8 (3 facilities), mild 2.5 (12), temperate 4.0 (14), hot humid 6.0 (13), and hot arid 8.0 (6). Han et al. estimate daily peaking factors of roughly 3 to 10 across facilities and service areas, including a lower-bound estimate of about 2.2 for a cooling-tower facility in Phoenix and about 6.5 for a dry-cooled facility with evaporative assist in West Des Moines [3]. Our values span a similar range, but they are

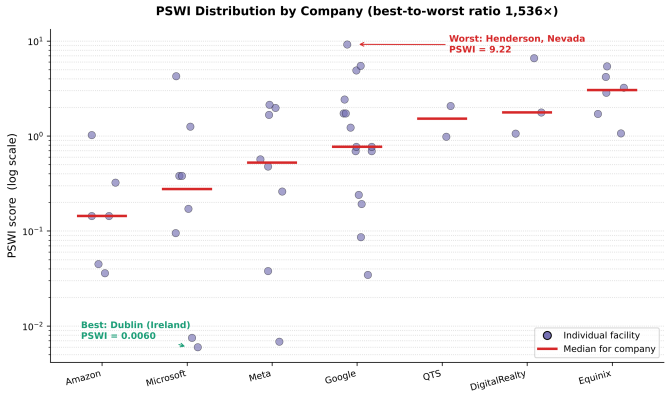


Fig. 1. PSWI by facility and operator (log scale). The lowest (Microsoft, Dublin; 0.006) and highest (Google, Henderson, NV; 9.22) differ by a factor of about 1,500.

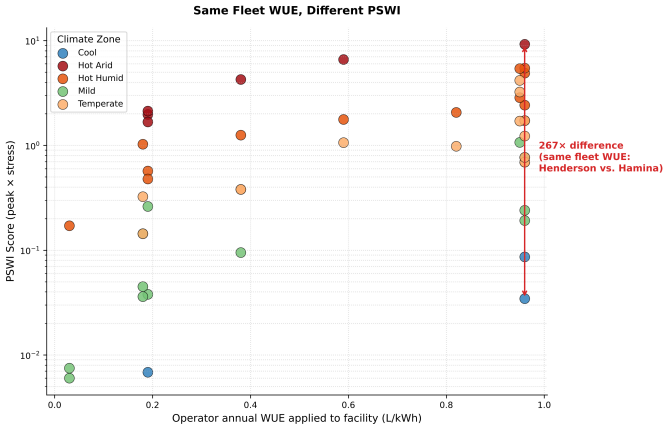


Fig. 2. PSWI versus the operator WUE applied to each facility (log scale), colored by climate class. Vertical stacks are facilities of one operator sharing a fleet or regional WUE. Google's Henderson and Hamina facilities differ 267-fold at identical WUE.

assumptions rather than measurements. In particular, the hot-arid value of 8.0 lies toward the upper end of Han et al.'s range and is not conservative for those sites. Because hot climates are often water-stressed, β and BWS are correlated; Section VI-E removes β entirely to test whether this drives our results.

E. Reproducibility and Data Provenance

All results can be reproduced from the public repository with three Python scripts (`pswi_calculator.py`, `sensitivity.py`, and `visualize.py`) using `pandas`, `numpy`, `scipy`, and `matplotlib`. For every facility, `datacenters.csv` records the WUE value, reporting year, basis (fleet, regional, or derived), and source. An earlier working version of this dataset assigned facility-specific WUE values to Google and Microsoft facilities that do not appear in those operators' published reports, and included five facilities that we could not verify or that did not meet the inclusion criteria. This version replaces all such values with published or published-derived figures; every change is logged in `dataset_changelog_v3.csv`.

VI. Results

A. Distribution of PSWI

PSWI ranges from 0.006 (Microsoft, Dublin) to 9.22 (Google, Henderson, Nevada), a ratio of about 1,500 (Fig. 1). The median is 0.88, with an interquartile range of 0.19 to 2.00. The range reflects both location and the operators' differing WUE values, which span 0.03 to 0.96 L/kWh.

B. Same Fleet WUE, Different PSWI

All 15 Google facilities carry the same derived fleet WUE (0.96 L/kWh), so differences in their PSWI arise entirely from location (Fig. 2). Henderson, Nevada ($\beta = 8.0$, BWS $= 1.20$; PSWI $= 9.22$) and Hamina, Finland ($\beta = 1.8$, BWS $= 0.02$; PSWI $= 0.035$) differ by a factor of 267. Of this, a factor of 60 comes from BWS and a factor of 4.4 from the assumed peaking factor. Lenoir, North Carolina ($\beta = 4.0$, BWS $= 0.18$; PSWI $= 0.69$) and Henderson differ 13-fold. Meta's facilities, also assigned a single fleet WUE, span a factor of 311, from Luleå, Sweden to Mesa, Arizona.

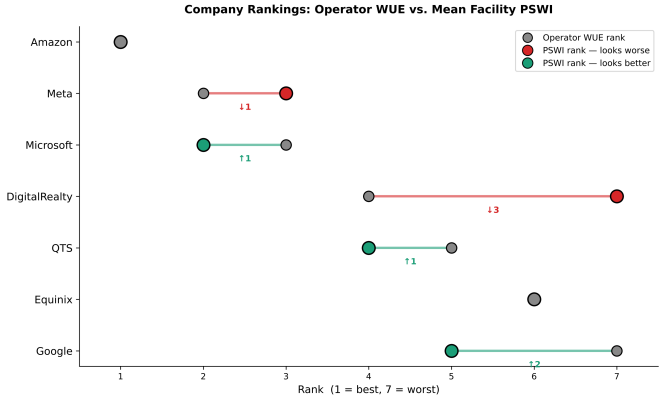


Fig. 3. Company ranks by operator WUE (gray) and by unweighted mean facility PSWI (colored). Shifts are illustrative: bootstrap intervals for all of them include zero (Fig. 4).

This is not evidence about the efficiency of individual facilities, which no operator in our sample discloses. It shows how much location-specific burden a single fleet-level number can conceal.

C. Company Rankings

Figure 3 compares company ranks by operator WUE with ranks by the unweighted mean PSWI of each company's facilities. The two orderings are related but distinct (Spearman $\rho = 0.71$). Digital Realty falls from fourth to seventh, driven by its Phoenix-area facility in Chandler, Arizona (PSWI $= 6.61$). Google rises from seventh to fifth because many of its facilities sit in low-stress catchments. Microsoft and QTS each rise one position and Meta falls one.

A within-operator bootstrap (Fig. 4) shows that none of these shifts is distinguishable from zero at the 95% level (Google: mean $+2.0$, interval $[0, +4]$; Digital Realty: mean

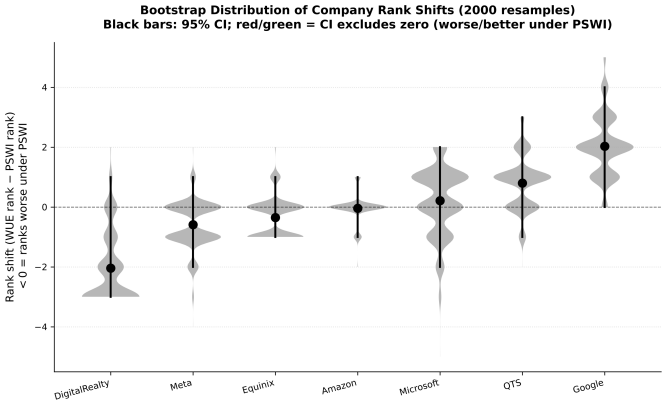


Fig. 4. Within-operator bootstrap (2,000 resamples) of company rank shifts, defined as WUE rank minus PSWI rank (negative values mean a company ranks worse under PSWI). Bars show 95% intervals; all include zero.

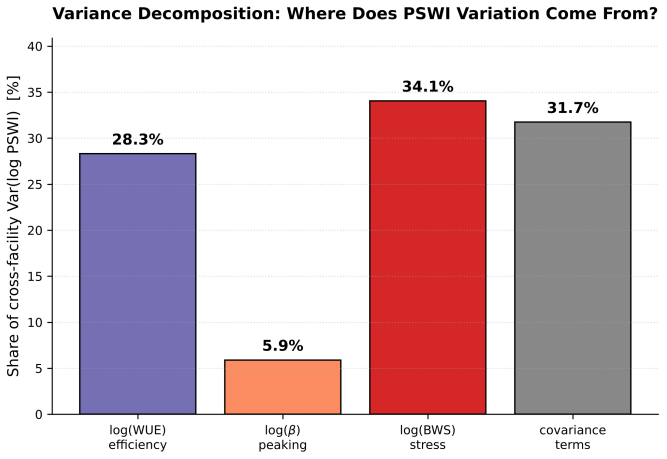


Fig. 5. Decomposition of $\mathrm{Var}(\log \mathrm{PSWI})$ across 48 facilities (Eq. 3). Most of the covariance share comes from the β–BWS term (23.9 of 31.7 points), reflecting the climate-based assignment of β.

-2.0, interval $[-3, +1]$). Company means are not weighted by capacity and depend on which facilities are sampled. We therefore treat company-level shifts as illustrative and the facility-level results as primary.

D. Variance Decomposition

Applying Eq. (3) across the 48 facilities (Fig. 5), the variance of $\log \mathrm{BWS}$ accounts for 34.1% of $\mathrm{Var}(\log \mathrm{PSWI})$, $\log \mathrm{WUE}$ for 28.3%, $\log \beta$ for 5.9%, and covariance terms for 31.7%. Nearly all of the covariance comes from the β–BWS pair (23.9 points; correlation 0.84 in logs), with WUE–BWS contributing 5.8 points and WUE–β 2.0. The β–BWS covariance is largely a consequence of assigning β by climate, and should not be read as independent evidence that peak demand and stress compound. The WUE share reflects differences between operators' published values, not differences between facilities of one operator.

E. Sensitivity to the Peaking Factor

Uniform rescaling of the β table leaves rankings unchanged by construction (Fig. 6a). Independent per-facility perturbations of β drawn uniformly within $\pm 50\%$ (1,000 runs) give a mean Spearman ρ of 0.978 between baseline and perturbed facility rankings (5th percentile 0.969), and retain on average 8.7 of the 10 highest-PSWI facilities (Fig. 6b).

The stricter test removes the peaking factor entirely ($\beta \equiv 1$, so that $\mathrm{PSWI} \propto \mathrm{WUE} \cdot \mathrm{BWS}$). Facility rankings then correlate at $\rho = 0.963$ with the main specification, and 9 of the 10 highest-PSWI facilities are unchanged. Company rankings correlate at $\rho = 0.964$, and no company's shift changes direction (Fig. 7). The peaking factor thus adds the peak-demand dimension to the magnitude of PSWI but has little effect on rankings in this sample, which are driven mainly by BWS. Measured, facility-specific peaking factors could change this.

F. Scope 2 Sensitivity

To test whether omitting indirect water changes the picture, we add 3.14 L/kWh, the U.S. average electricity water-consumption intensity used by Li et al. [10], to every facility's WUE before applying β and BWS. Facility rankings correlate at $\rho = 0.88$ with the main specification, 6 of the 10 highest-PSWI facilities are retained, and the mean absolute rank change is 5.2 positions. This is a crude stress test: Scope 2 water is drawn at power plants, often in other basins with different peaking behavior, and a U.S. average is applied to non-U.S. sites. A proper treatment would weight indirect water by the stress of the generating basin, as AWI does [8].

G. Geographic Concentration

Of the 10 highest-PSWI facilities (Fig. 8), five are in the U.S. West (Henderson, Nevada; Chandler and Goodyear, Arizona; El Segundo and San Jose, California), three are in Singapore and Taiwan, and two are in Texas and Georgia. Of the 10 lowest, seven are in Northern and Western Europe (Hamina, Finland; Luleå, Sweden; Eemshaven and Amsterdam, Netherlands; and three in the Dublin area), two are in the U.S. Pacific Northwest (Boardman, Oregon; Quincy, Washington), and one is in Ohio. Several high-PSWI locations are established data center markets, so siting decisions there carry disproportionate water-stress consequences.

H. Disclosure Audit

Figure 9 summarizes what the documents cited in our dataset disclose for each operator, with xAI shown for reference. Six of the seven sampled operators publish a fleet-level WUE; Google does not, although one can be derived from its published totals. Only Microsoft publishes WUE below the fleet level, and only by region. Google and Meta publish facility-level water volumes but not the facility energy needed to compute WUE. We found no peak-day or monthly water data from any operator, and no WUE from xAI. Because no operator publishes facility-level WUE or peak data, no outside party can compute a fully facility-specific PSWI for any facility in our sample.

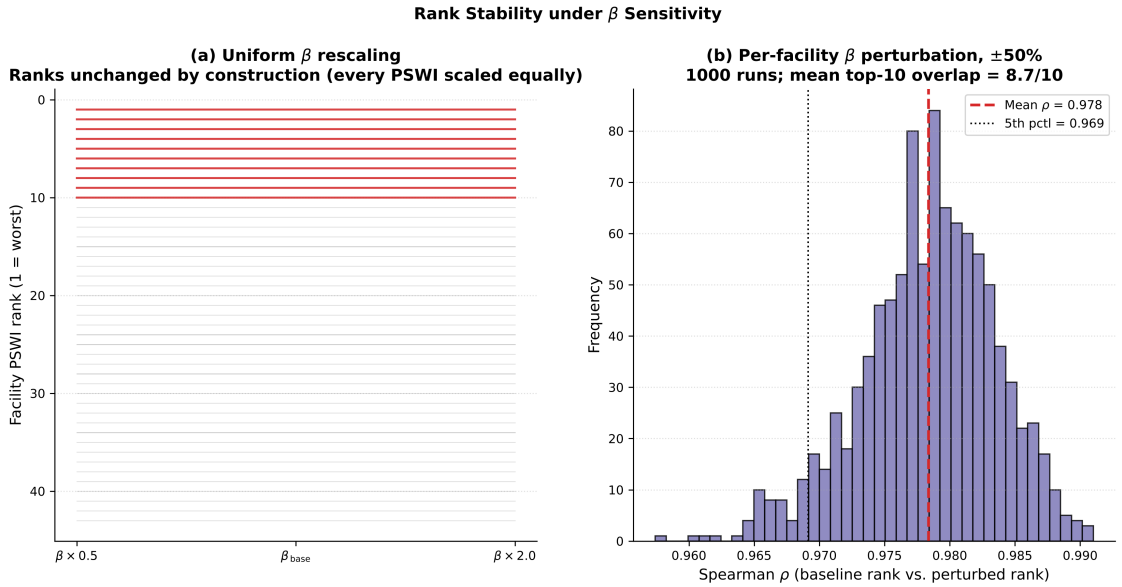


Fig. 6. (a) Uniform rescaling of β leaves ranks unchanged by construction. (b) Independent per-facility perturbation of β by up to $\pm 50\%$ (1,000 runs): mean Spearman $\rho = 0.978$ between baseline and perturbed facility rankings.

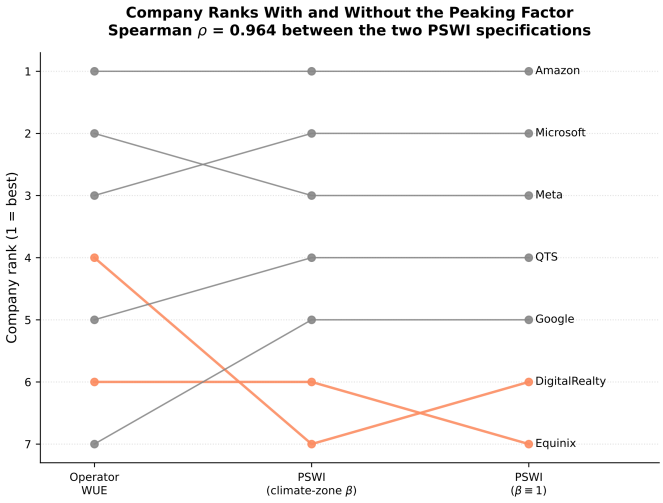


Fig. 7. Company ranks under operator WUE, PSWI with climate-class β, and PSWI with $\beta \equiv 1$. Rankings under the two PSWI specifications correlate at $\rho = 0.964$.

VII. Counterfactual Analysis and Policy Implications

A. Idealized Reallocation Counterfactual

a) Setup: Each facility starts with a workload of one unit. A fraction f of total workload is removed from the highest-PSWI third of facilities, in proportion to their PSWI, and reassigned to the lowest-PSWI third, in inverse proportion to their PSWI. In each of 500 Monte Carlo draws, WUE, BWS, and β are perturbed independently by normal relative errors with standard deviations of 15%, 25%, and 30% (β floored at 1). These uncertainty levels are assumptions chosen to be generous, not estimates from a published error model.

b) Results: At $f \approx 10\%$, aggregate PSWI falls by a median of 35.1% (95% interval 20.9–46.1%); at 20%, by 57.5% (48.1–63.9%); and at 30%, by 67.3% (60.1–72.7%) (Fig. 10).

c) Scope of the claim: This is an accounting exercise. Because workload moves from high- to low-PSWI facilities, aggregate PSWI must fall; what the simulation adds is the size and concavity of the reduction, which reflect how concentrated PSWI is in a small tail of facilities. It ignores latency, data-residency, and capacity constraints, the carbon intensity of receiving grids, and the fact that operators move workloads among their own facilities rather than across a pooled fleet [21], [22]. The figures are upper bounds under frictionless reallocation, not forecasts of what any policy would achieve.

B. Disclosure Framework

Three disclosures would allow any party to compute a facility-level PSWI from public data: (i) *facility-level annual WUE*, together with water withdrawal, water consumption, and IT energy, without aggregation across facilities; (ii) *monthly water use for each facility*, from which an observed peaking factor can be computed (the ratio of peak to average monthly use is a lower bound on the daily peaking factor [3]); and (iii) *source-resolved water inputs*, distinguishing potable from non-potable water and utility supply from self-supply. Requirements could phase in by facility size, beginning with the largest facilities (for example, above 50 MW of IT load) and extending to smaller ones later, with limited third-party assurance preceding reasonable assurance.

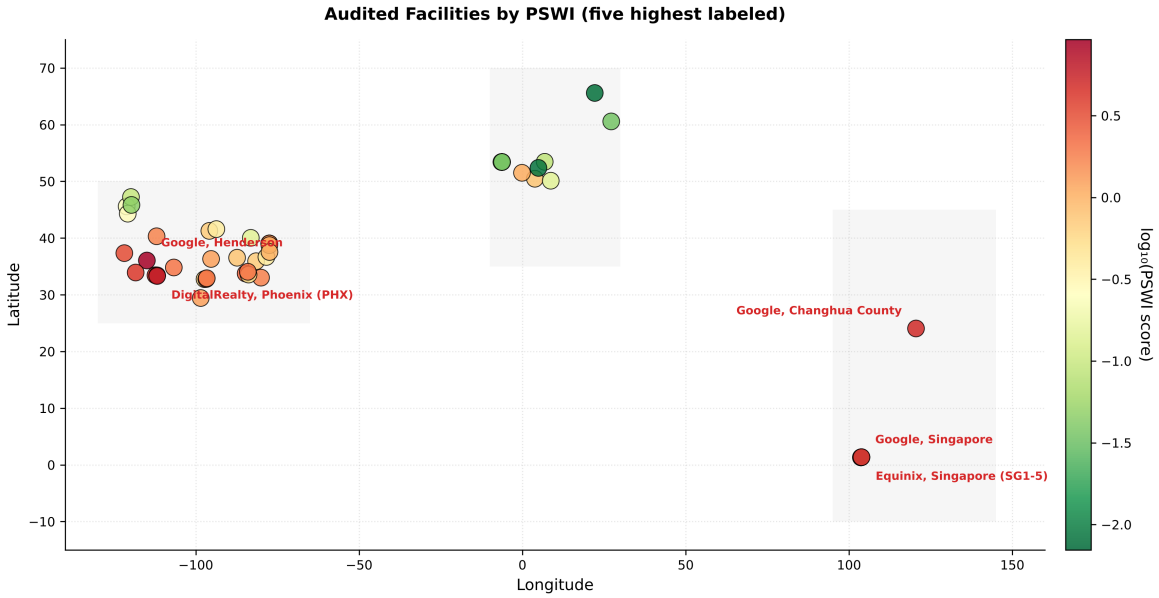


Fig. 8. Audited facilities colored by $\log_{10}$(PSWI); the five highest are labeled. High values cluster in the U.S. West and in Singapore and Taiwan; Northern and Western Europe are low.

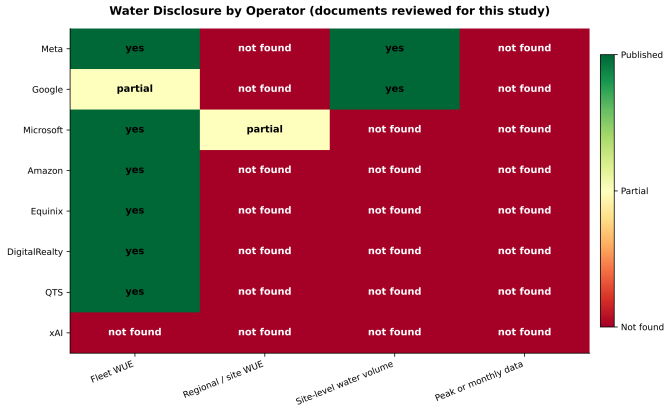


Fig. 9. Water disclosure by operator in the documents reviewed for this study. "Not found" means absent from those documents. No operator publishes peak or monthly water data.

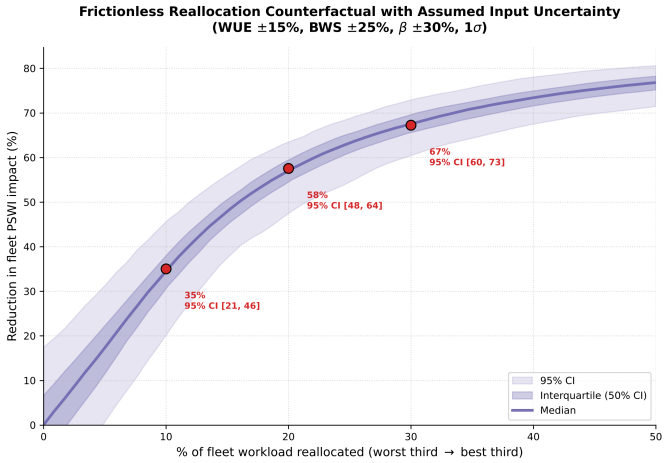


Fig. 10. Idealized, frictionless reallocation from the highest- to the lowest-PSWI third of facilities with assumed input uncertainty (WUE ±15%, BWS ±25%, β ±30%, 1σ; 500 runs). At 10% reallocation the median reduction in aggregate PSWI is 35% (95% interval 21–46%).

C. Lessons from Federal Climate Disclosure

The SEC's climate disclosure rules, adopted in March 2024 [32], used a similar architecture within Regulation S-K [33]: phased compliance beginning with the largest filers, limited assurance before reasonable assurance for greenhouse gas emissions, and materiality qualifiers. Their history is also a caution. The SEC stayed the rules in April 2024 pending litigation, voted in March 2025 to stop defending them, and in May 2026 proposed rescinding them, arguing that it lacked statutory authority to adopt them [34]. Securities disclosure also reaches only publicly traded issuers, leaving privately held operators uncovered. We therefore borrow the architecture of these rules, not their legal vehicle.

D. Implementation through Water Governance

A more direct route runs through existing water governance. Utilities and local governments already decide whether and how to serve large new customers, and Han et al. show that available peak capacity is increasingly the binding constraint on those decisions [3]. States separately regulate many withdrawals through water-rights and groundwater permitting. Requiring PSWI inputs as a condition of new large service connections or withdrawal permits would place disclosure where water decisions are made, and would cover private as well as public operators. Whether existing authority suffices varies by state; assessing that is beyond the scope of this paper.

VIII. Limitations and Future Work

a) No facility-level WUE: All within-operator variation in our dataset comes from location, so real efficiency differences between facilities—for example, a dry-cooled site in a hot climate—are invisible. Operators also define WUE differently and report for different years. Microsoft's Asia-Pacific WUE, for example, was 0.03 L/kWh in FY24 but 0.25 in FY25 [5], which would raise the PSWI of its Singapore facility roughly eightfold.

b) Assumed peaking factors: β is assigned by climate class and is correlated with BWS. Monthly facility-level water data would allow measured peaking factors.

c) Water stress inputs: BWS values are annual, rounded, and assigned manually by metropolitan area. A geospatial join of facility coordinates to Aqueduct catchments, and the use of Aqueduct's monthly stress values to capture summer scarcity, are the highest-priority extensions.

d) Sample: The sample is purposive, with few facilities for some operators, and company means are not weighted by capacity. Coverage of Asia, the Middle East, and undisclosing operators is limited or absent.

e) Scope 2 and validation: Indirect water is excluded from the main specification. PSWI is a screening indicator, not a measure of harm; validating it against observed water-system stress events is a natural next step.

IX. Conclusion

Annual, fleet-averaged WUE conceals two features that matter for host communities: when water is drawn and how scarce it is where it is drawn. PSWI combines both in a single facility-level index. Applied to 48 facilities using only published or published-derived inputs, it spans a factor of about 1,500, and a single operator's facilities differ 267-fold sharing one fleet WUE. These results are driven mainly by location stress and are robust to removing the assumed peaking factor. Company-level ranking shifts are suggestive but not statistically distinguishable from zero. Because no operator publishes facility-level WUE or peak water data, the most useful next step is disclosure: facility-level WUE, monthly water use, and source-resolved inputs, implemented through the water governance that already decides where large users connect. All data and code are publicly released.

Data and Code Availability

The dataset, analysis code, figures, and change log are available at https://github.com/hrihaan19/pswi-data-center-audit. Code is released under the MIT License and data under CC BY 4.0. Aqueduct 4.0 data are published by the World Resources Institute under CC BY 4.0.

Use of AI Tools

The author used Claude, a large language model developed by Anthropic, as an assistant throughout this project, including drafting and editing text, writing and debugging analysis code, and checking data values against the public sources cited. All values, code, and results were checked against those sources and the released code. The author takes full responsibility for the content of this paper.

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